IMDb Movies Data Analysis

1. Project Summary

This project analyzes 600 IMDb movie records by combining movie information and business data. The dataset was cleaned, merged, and explored to understand genres, ratings, runtime, gross income, director performance, and hit/flop trends.

2. Project Objectives

To examine data quality, handle missing values, merge datasets, analyze movie characteristics, and identify key trends through exploratory data analysis.

3. Tools and Techniques Used

Python, Pandas, NumPy, Matplotlib, and Seaborn were used for data cleaning, analysis, visualization, and statistical calculations.

4. Data Preparation

The datasets were inspected for missing values and duplicates. Selected numerical values were filled using averages, text fields were handled appropriately, and both datasets were merged into a single analysis table.

5. Exploratory Analysis

Genre distribution, IMDb ratings, runtime, gross income, director performance, and hit/flop categories were analyzed using summary statistics and visualizations.

6. Main Findings

Thriller and Drama were the most common genres. The average IMDb rating was around 7.09, while Horror had the highest average genre rating. Northern World recorded the highest gross income, and the relationship between ratings and income was weak. Brian Taylor appeared most frequently among directors.

7. Limitations

A small number of missing values remained in some columns. Results are based only on the given dataset and should not be generalized to all movies.

8. Presentation Summary

The project demonstrates a complete workflow of data cleaning, merging, visualization, and analysis using Python. The findings provide insights into movie performance, genres, ratings, and revenues.

9. Conclusion

The analysis successfully identified important patterns in the IMDb dataset. It highlights how data analytics can be used to extract meaningful insights and support decision-making from large datasets.